claude-windows / 985cc286-9e1e-4217-87c3-2d418bc4fa9c

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\985cc286-9e1e-4217-87c3-2d418bc4fa9c\subagents\workflows\wf_2cab3781-897\agent-a1b9a52415a04a2ee.jsonl`
- SHA-256: `e43b288318b309da950b13b4475971d8e6a0288e06c0a5450b0bc15c9e331226`
- Source modified: `2026-06-14T07:04:38+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_2cab3781-897`
- session_id: `985cc286-9e1e-4217-87c3-2d418bc4fa9c`

Transcript

1. user (2026-06-14T07:01:42.963Z)

Restyle the existing PDF report generator at scratch/quality_report.py in the badminton-highlight-indexer repo (cwd) to a CONSERVATIVE, academic-journal-acceptable look so a strict professor will NOT reject it for "weird colors / formatting". Make SURGICAL edits to the style constants, font registration, the table-builder, and the title-page/banner function ? DO NOT change the report's text content or numbers. Apply ALL of these:

1. BODY FONT = serif. Register Times New Roman: C:/Windows/Fonts/times.ttf (regular), timesbd.ttf (bold), timesi.ttf (italic), timesbi.ttf (bolditalic) via TTFont + registerFontFamily; use it for body, abstract, captions. (Fall back to the existing Arial only if Times is missing.) Body ~10pt, leading ~13-14, justified.
2. COLORS = minimal & muted. Keep ONE accent only: dark navy #1F3A5F (or the existing #13294B) for section headings and thin rules. DELETE the crimson accent #C8102E entirely (subtitle and any red strip -> black or dark grey). DELETE all conditional GREEN/RED number coloring in tables -> ALL numbers plain black; the +/- sign already conveys direction.
3. TABLES = booktabs style. No heavy colored header fill and no full vertical grid. Use: a thin top rule, a header row in BOLD BLACK on white (or a very light grey #F0F0F0 fill at most), a thin rule under the header, horizontal hairlines only (or none between body rows), and a thin bottom rule. Replace any bright header background (#1F4E6B with white text) with this restrained style. Keep alternating row shading OFF or extremely faint (#F7F7F7).
4. TITLE PAGE = clean academic, NO colored banner block. Remove the navy banner rectangle + red strip. Use a centered title (serif bold, dark navy or black), a black italic subtitle, the affiliation line, and the abstract below ? like a real paper's first page. Page numbers + a thin running header (grey, small) on pages 2+ are fine.
5. FIGURE: make the matplotlib bar chart muted ? grey/navy bars (e.g. '#1F3A5F' and '#9AA7B4'), black axis labels, a thin dashed grey floor line; no bright/saturated colors. Save at >=150 dpi so it's crisp.

Then RUN it: `python scratch/quality_report.py`. Confirm scratch/Rally_Detection_Quality_Report.pdf regenerated (ls -la). Then render ALL pages to PNG for review: `python -c "import fitz; d=fitz.open('scratch/Rally_Detection_Quality_Report.pdf'); [d[i].get_pixmap(dpi=120).save(f'scratch/_rev_p{i+1}.png') for i in range(len(d))]"`. Return: confirmation, PDF size + page count, and a 2-line summary of what you changed.

2. assistant (2026-06-14T07:01:45.334Z)

I'll start by reading the existing report generator to understand its structure.

3. user (2026-06-14T07:01:46.398Z)

```
"""Generate a paper-style quality report PDF for the rally-detection system.

Self-contained: uses reportlab (Platypus) + matplotlib. Produces a polished
multi-page PDF at scratch/Rally_Detection_Quality_Report.pdf. All numbers are
verified facts pulled from the codebase (docs/QUALITY_ITERATIONS.md and the
```

```python
eval_baselines/*.json leaderboard records); nothing here is invented.

Run:  python scratch/quality_report.py
"""

import os

import matplotlib

matplotlib.use("Agg")
import matplotlib.pyplot as plt
from reportlab.lib import colors
from reportlab.lib.enums import TA_CENTER, TA_JUSTIFY, TA_LEFT
from reportlab.lib.pagesizes import A4
from reportlab.lib.styles import ParagraphStyle, getSampleStyleSheet
from reportlab.lib.units import cm, mm
from reportlab.pdfbase import pdfmetrics
from reportlab.pdfbase.ttfonts import TTFont
from reportlab.platypus import (
    BaseDocTemplate,
    Frame,
    Image,
    NextPageTemplate,
    PageBreak,
    PageTemplate,
    Paragraph,
    Spacer,
    Table,
    TableStyle,
)

HERE = os.path.dirname(os.path.abspath(__file__))
PDF_PATH = os.path.join(HERE, "Rally_Detection_Quality_Report.pdf")
CHART_PATH = os.path.join(HERE, "_fusion_ablation_chart.png")

# ---------------------------------------------------------------------------
# Fonts: register a Unicode TTF so glyphs like the delta, >=, +/-, x render.
# ---------------------------------------------------------------------------
FONT_REGULAR = "Helvetica"
FONT_BOLD = "Helvetica-Bold"
FONT_ITALIC = "Helvetica-Oblique"

_arial_candidates = [
    os.path.join(HERE, "arial.ttf"),
    r"C:/Windows/Fonts/arial.ttf",
]
_arialbd_candidates = [
    r"C:/Windows/Fonts/arialbd.ttf",
    os.path.join(HERE, "arialbd.ttf"),
]
_ariali_candidates = [
    r"C:/Windows/Fonts/ariali.ttf",
    os.path.join(HERE, "ariali.ttf"),
]


def _first_existing(paths):
    for p in paths:
        if os.path.exists(p):
            return p
    return None


_reg = _first_existing(_arial_candidates)
if _reg:
```

```
    try:
        pdfmetrics.registerFont(TTFont("UI", _reg))
        FONT_REGULAR = "UI"
        bd = _first_existing(_arialbd_candidates)
        it = _first_existing(_ariali_candidates)
        pdfmetrics.registerFont(TTFont("UI-Bold", bd if bd else _reg))
        pdfmetrics.registerFont(TTFont("UI-Italic", it if it else _reg))
        FONT_BOLD = "UI-Bold"
        FONT_ITALIC = "UI-Italic"
        # Map the family so <b>/<i> inline tags resolve to the TTF variants.
        pdfmetrics.registerFontFamily(
            "UI", normal="UI", bold="UI-Bold", italic="UI-Italic", boldItalic="UI-Bold"
        )
    except Exception as exc:  # pragma: no cover - defensive
        print("Font registration fell back to Helvetica:", exc)
        FONT_REGULAR, FONT_BOLD, FONT_ITALIC = (
            "Helvetica",
            "Helvetica-Bold",
            "Helvetica-Oblique",
        )

UNICODE_OK = FONT_REGULAR == "UI"
DELTA = "?" if UNICODE_OK else "Delta"  # Greek capital delta
GE = "?" if UNICODE_OK else ">="
PM = "±" if UNICODE_OK else "+/-"
TIMES = "×" if UNICODE_OK else "x"
ARROW = "?" if UNICODE_OK else "->"

# Palette
NAVY = colors.HexColor("#13294B")
STEEL = colors.HexColor("#2C5F8A")
ACCENT = colors.HexColor("#C8102E")
LIGHT = colors.HexColor("#EAF0F6")
MIDGREY = colors.HexColor("#5A6675")
HEADER_BG = colors.HexColor("#1F4E6B")
ROW_ALT = colors.HexColor("#F2F6FA")
GREEN = colors.HexColor("#1B7A3D")

# ---------------------------------------------------------------------------
# Styles
# ---------------------------------------------------------------------------
styles = getSampleStyleSheet()


def style(name, **kw):
    base = ParagraphStyle(name, parent=styles["Normal"], **kw)
    return base


BODY = style(
    "Body",
    fontName=FONT_REGULAR,
    fontSize=9.5,
    leading=14,
    alignment=TA_JUSTIFY,
    spaceAfter=6,
    textColor=colors.HexColor("#1A1A1A"),
)
BODY_TIGHT = style(
    "BodyTight",
    fontName=FONT_REGULAR,
    fontSize=9.5,
    leading=13.5,
    alignment=TA_JUSTIFY,
    spaceAfter=3,
```

```
    )
    H1 = style(
        "H1",
        fontName=FONT_BOLD,
        fontSize=14,
        leading=18,
        textColor=NAVY,
        spaceBefore=14,
        spaceAfter=6,
    )
    H2 = style(
        "H2",
        fontName=FONT_BOLD,
        fontSize=11,
        leading=15,
        textColor=STEEL,
        spaceBefore=8,
        spaceAfter=4,
    )
    ABSTRACT = style(
        "Abstract",
        fontName=FONT_ITALIC,
        fontSize=9.5,
        leading=14,
        alignment=TA_JUSTIFY,
        textColor=colors.HexColor("#222222"),
        leftIndent=10,
        rightIndent=10,
        spaceAfter=6,
    )
    CAPTION = style(
        "Caption",
        fontName=FONT_ITALIC,
        fontSize=8,
        leading=11,
        alignment=TA_CENTER,
        textColor=MIDGREY,
        spaceBefore=2,
        spaceAfter=8,
    )
    TITLE = style(
        "Title2",
        fontName=FONT_BOLD,
        fontSize=22,
        leading=27,
        alignment=TA_CENTER,
        textColor=NAVY,
        spaceAfter=4,
    )
    SUBTITLE = style(
        "Subtitle",
        fontName=FONT_ITALIC,
        fontSize=11.5,
        leading=15,
        alignment=TA_CENTER,
        textColor=ACCENT,
        spaceAfter=2,
    )
    AUTHORLINE = style(
        "Author",
        fontName=FONT_REGULAR,
        fontSize=9,
        leading=13,
        alignment=TA_CENTER,
        textColor=MIDGREY,
```

```
    spaceAfter=14,
)
TBL_CELL = style("TblCell", fontName=FONT_REGULAR, fontSize=8.3, leading=10.5)
TBL_HEAD = style(
    "TblHead",
    fontName=FONT_BOLD,
    fontSize=8.4,
    leading=10.5,
    textColor=colors.white,
)
KEYBOX = style(
    "KeyBox",
    fontName=FONT_REGULAR,
    fontSize=9,
    leading=13,
    textColor=NAVY,
    leftIndent=6,
    rightIndent=6,
    spaceAfter=2,
)


def P(text, st=BODY):
    return Paragraph(text, st)


# ---------------------------------------------------------------------------
# Matplotlib chart: fusion ablation (uses full-precision JSON floats)
# ---------------------------------------------------------------------------
def build_chart():
    # Full-precision values from eval_baselines/experiment_leaderboard.json
    labels = ["shuttle-only", "+person (P3)", "+person+audio (P4)"]
    f1 = [0.3068611057861412, 0.2863398846047612, 0.3655826552797436]
    # 95% bootstrap CI half-widths are only defined for the incremental
    # deltas, not for absolute F1; we annotate deltas as text instead.
    bar_colors = ["#2C5F8A", "#9AA7B5", "#1B7A3D"]

    fig, ax = plt.subplots(figsize=(6.6, 3.5), dpi=150)
    x = range(len(labels))
    bars = ax.bar(x, f1, color=bar_colors, width=0.58, edgecolor="#1A2A38",
linewidth=0.6)

    for i, (b, v) in enumerate(zip(bars, f1)):
        ax.text(
            b.get_x() + b.get_width() / 2,
            v + 0.006,
            f"{v:.3f}",
            ha="center",
            va="bottom",
            fontsize=10,
            fontweight="bold",
            color="#13294B",
        )

    # delta annotations between consecutive bars
    ax.annotate(
        "?F1 = -0.021\n(CI spans 0)",
        xy=(0.5, 0.33),
        ha="center",
        fontsize=8,
        color="#C8102E",
    )
    ax.annotate(
        "?F1 = +0.079\n(CI spans 0)",
        xy=(1.5, 0.39),
```

```python
            ha="center",
            fontsize=8,
            color="#1B7A3D",
        )

        # reference line: heuristic LOVO floor
        ax.axhline(0.480, color="#C8102E", linestyle="--", linewidth=1.0, alpha=0.8)
        ax.text(
            2.42, 0.487, "heuristic LOVO floor 0.480", ha="right", va="bottom",
            fontsize=7.5, color="#C8102E",
        )

        ax.set_xticks(list(x))
        ax.set_xticklabels(labels, fontsize=9)
        ax.set_ylabel("LOVO F1  (IoU >= 0.5, n=6)", fontsize=9.5)
        ax.set_ylim(0, 0.55)
        ax.set_title(
            "Fusion ablation: incremental cue contribution (honest LOVO)",
            fontsize=10.5,
            fontweight="bold",
            color="#13294B",
        )
        ax.spines["top"].set_visible(False)
        ax.spines["right"].set_visible(False)
        ax.yaxis.grid(True, linestyle=":", alpha=0.4)
        ax.set_axisbelow(True)
        fig.tight_layout()
        fig.savefig(CHART_PATH, bbox_inches="tight", facecolor="white")
        plt.close(fig)


    # ---------------------------------------------------------------------------
    # Table helper
    # ---------------------------------------------------------------------------
    def make_table(data, col_widths, header_rows=1, align_centers=None, font_size=8.3):
        """data: list of rows; row[i] is either a str or a Paragraph."""
        wrapped = []
        for r_idx, row in enumerate(data):
            new_row = []
            for c_idx, cell in enumerate(row):
                if isinstance(cell, Paragraph):
                    new_row.append(cell)
                else:
                    st = TBL_HEAD if r_idx < header_rows else TBL_CELL
                    new_row.append(Paragraph(str(cell), st))
            wrapped.append(new_row)

        t = Table(wrapped, colWidths=col_widths, repeatRows=header_rows)
        cmds = [
            ("BACKGROUND", (0, 0), (-1, header_rows - 1), HEADER_BG),
            ("TEXTCOLOR", (0, 0), (-1, header_rows - 1), colors.white),
            ("VALIGN", (0, 0), (-1, -1), "MIDDLE"),
            ("TOPPADDING", (0, 0), (-1, -1), 4),
            ("BOTTOMPADDING", (0, 0), (-1, -1), 4),
            ("LEFTPADDING", (0, 0), (-1, -1), 5),
            ("RIGHTPADDING", (0, 0), (-1, -1), 5),
            ("GRID", (0, 0), (-1, -1), 0.4, colors.HexColor("#B8C4D0")),
            ("LINEBELOW", (0, header_rows - 1), (-1, header_rows - 1), 1.0, NAVY),
            ("ROWBACKGROUNDS", (0, header_rows), (-1, -1), [colors.white, ROW_ALT]),
        ]
        if align_centers:
            for c in align_centers:
                cmds.append(("ALIGN", (c, 0), (c, -1), "CENTER"))
        t.setStyle(TableStyle(cmds))
        return t
```

```


# ---------------------------------------------------------------------------
# Page decoration: header rule + footer page number
# ---------------------------------------------------------------------------
def _footer(canvas, doc):
    w, h = A4
    canvas.saveState()
    canvas.setStrokeColor(colors.HexColor("#C6D0DA"))
    canvas.setLineWidth(0.6)
    canvas.line(2 * cm, 1.4 * cm, w - 2 * cm, 1.4 * cm)
    canvas.setFont(FONT_REGULAR, 8)
    canvas.setFillColor(MIDGREY)
    canvas.drawCentredString(w / 2.0, 1.0 * cm, f"{doc.page}")
    canvas.drawString(2 * cm, 1.0 * cm, "badminton-highlight-indexer")
    canvas.drawRightString(w - 2 * cm, 1.0 * cm, "Internal · honest measured results only")
    canvas.restoreState()


def on_page(canvas, doc):
    canvas.saveState()
    w, h = A4
    # top running rule + header text
    canvas.setStrokeColor(NAVY)
    canvas.setLineWidth(0.8)
    canvas.line(2 * cm, h - 1.4 * cm, w - 2 * cm, h - 1.4 * cm)
    canvas.setFont(FONT_ITALIC, 7.5)
    canvas.setFillColor(MIDGREY)
    canvas.drawString(2 * cm, h - 1.25 * cm, "Local, Cheap, AI Rally Detection for Racket Sports")
    canvas.drawRightString(
        w - 2 * cm, h - 1.25 * cm, "Project Quality Report ? toward a paper"
    )
    canvas.restoreState()
    _footer(canvas, doc)


def on_first_page(canvas, doc):
    # Title page gets a colored banner instead of the running top header.
    canvas.saveState()
    w, h = A4
    canvas.setFillColor(NAVY)
    canvas.rect(0, h - 4.4 * cm, w, 4.4 * cm, fill=1, stroke=0)
    canvas.setFillColor(ACCENT)
    canvas.rect(0, h - 4.5 * cm, w, 0.1 * cm, fill=1, stroke=0)
    # Banner caption in the navy strip (white, so it reads on the banner).
    canvas.setFillColor(colors.white)
    canvas.setFont(FONT_ITALIC, 9)
    canvas.drawCentredString(
        w / 2.0, h - 1.0 * cm, "Project Quality Report  ·  honest measured results only"
    )
    canvas.restoreState()
    # Only the bottom footer/page number on the title page (no top running rule).
    _footer(canvas, doc)


# ---------------------------------------------------------------------------
# Build the story
# ---------------------------------------------------------------------------
def build_story():
    story = []

    # --- Title block (sits below the banner) ---
    story.append(Spacer(1, 0.2 * cm))
```

```
        story.append(Paragraph("Local, Cheap, AI Rally Detection<br/>for Racket Sports", TITLE))
        story.append(Paragraph("Project report &mdash; toward a paper", SUBTITLE))
        story.append(
            Paragraph(
                "An honest empirical study of a shuttle-trajectory rally segmenter, "
                "its disciplined small-N promotion gate, and a fusion ablation",
                AUTHORLINE,
            )
        )

        # --- Abstract ---
        story.append(Paragraph("Abstract", H1))
        abstract = (
            "We present a fully local, low-cost rally-detection system for racket sports "
            "(badminton first) that segments a match video into rally <i>[start, end]</i> "
            "intervals. The system is built on a frozen shuttle-trajectory substrate (a WASB CNN "
            "that emits per-frame shuttle position), from which a velocity signal seeds "
            "recall-oriented candidate windows that a tiny pure-numpy logistic-regression scorer "
            "grades. Every additional cue enters as a confidence-weighted feature column that the "
            "scorer learns to weight &mdash; never as a hand-coded branch &mdash; and ships "
            "<b>default-OFF</b>, promoted to the served default only on a measured held-out lift. "
            "We evaluate against a small owned golden corpus of <b>n=6</b> labelled matches, treating "
            "the labelling vendor (Roora) as an interchangeable <i>oracle</i>: the contract is with the "
            "golden labels as trustable ground truth, not with their provenance, so evaluation rigor "
            "is decoupled from the vendor and the corpus scales by adding oracles. Using leave-one-"
            "video-out cross-validation with bootstrap 95% confidence intervals and a paired sign "
            "test, our first fused cue &mdash; a net-crossing gate (Gate A) &mdash; is honestly "
            f"<b>promoted</b> ({DELTA}F1 = +0.074, 6/6 matches, sign-test p = 0.031, CI [+0.042, +0.104]), "
            "cutting over-segmentation from 1.68 to 1.01. Two further cues (a person cue, then audio) "
            "do <i>not</i> clear the bar at n=6 (both CIs span 0) and stay default-OFF. We report these "
            "negative results deliberately as the honest ablation backbone, frame the architecture "
            "against standard prior art (temporal action localization / segmentation + late fusion), "
            "and isolate the genuinely novel piece: a report-first reconciler-as-linter paired with a "
            "disciplined small-N promotion gate."
        )
        story.append(Paragraph(abstract, ABSTRACT))

        # Key-result strip
        kr = [
            [P(f"<b>Gemini wrapper (cloud ref.)</b>", TBL_CELL), P("clean 0.93 / multi-court 0.66", TBL_CELL)],
            [P("<b>Gen-0 owned head (LOVO)</b>", TBL_CELL), P("clean 0.744 / multi-court 0.561", TBL_CELL)],
            [P("<b>Gate-A (PROMOTED)</b>", TBL_CELL), P(f"{DELTA}F1 +0.074, p=0.031, CI [+0.042,+0.104]", TBL_CELL)],
            [P("<b>Golden corpus</b>", TBL_CELL), P("n=6 matches (262 rallies), Proprietary, minors excluded", TBL_CELL)],
```

```
        ]
        kt = Table(kr, colWidths=[5.2 * cm, 11.3 * cm])
        kt.setStyle(
            TableStyle(
                [
                    ("BACKGROUND", (0, 0), (0, -1), LIGHT),
                    ("BOX", (0, 0), (-1, -1), 0.6, STEEL),
                    ("INNERGRID", (0, 0), (-1, -1), 0.4, colors.HexColor("#C6D0DA")),
                    ("VALIGN", (0, 0), (-1, -1), "MIDDLE"),
                    ("TOPPADDING", (0, 0), (-1, -1), 3),
                    ("BOTTOMPADDING", (0, 0), (-1, -1), 3),
                    ("LEFTPADDING", (0, 0), (-1, -1), 6),
                ]
            )
        )
        story.append(Spacer(1, 0.2 * cm))
        story.append(kt)
        story.append(Paragraph("Headline results at a glance (IoU 0.5 vs human golden labels).", CAPTION))

        # ================= 1. Introduction =================
        story.append(Paragraph("1  Introduction &amp; Problem", H1))
        story.append(
            P(
                "A rally is the unit of play in racket sports: the span from serve contact to the point "
                "ending. Indexing a long match video into its rallies is the substrate for everything a "
                "coach or player wants &mdash; highlight reels, &ldquo;show me the longest rally,&rdquo; "
                "per-rally statistics. Cloud vision-language models can do this well on clean footage, but "
                "they are expensive per video, non-deterministic, and (critically) <b>cannot be a training "
                "data source</b> for an owned model under their terms. Our constraint is the opposite end of "
                "the design space: a system that runs <b>locally, cheaply, and reproducibly</b> on a single "
                "consumer GPU, with paid LLMs allowed only as an optional inference fallback &mdash; never "
                "in the training loop."
            )
        )
        story.append(
            P(
                "Two failure modes dominate and motivate the entire method. <b>(1) Held-shuttle false "
                "positives:</b> a player holding or flicking the shuttle between points produces "
                "player-motion that the old velocity heuristic counted as a rally &mdash; the owner-observed "
                "precision bug. <b>(2) Over-segmentation:</b> a single true rally gets shredded into several "
                "short fragments. Crucially, the standard temporal-IoU metric is nearly <i>blind</i> to "
                "over-segmentation, so a method can look fine on IoU while destroying rally structure. Both "
                "failure modes are about the boundary between play and non-play, which is exactly what the "
                "fusion gates in this report target."
            )
        )

        # ================= 2. System & Method =================
        story.append(Paragraph("2  System &amp; Method", H1))
```

```
        story.append(Paragraph("2.1  A two-layer mental model", H2))
        story.append(
            P(
                "<b>Layer 1 &mdash; Detection (&ldquo;where is the shuttle this frame?&rdquo;).</b> A frozen "
                "WASB CNN emits a per-frame (x, y, visible) shuttle track. The shuttle is chosen as the "
                "camera-<i>invariant</i> primary signal because &ldquo;shuttle in flight&rdquo; is a near-"
                "synonym for &ldquo;a rally is happening,&rdquo; unlike player motion which fooled the old "
                "heuristic on broadcast pan cameras. <b>Layer 2 &mdash; Segmentation (&ldquo;when is a "
                "rally?&rdquo;).</b> From the per-frame track we compute a shuttle <i>velocity</i> signal, "
                "which seeds recall-oriented candidate windows; a learned scorer then grades each window "
                f"into final [start, end] rallies measured by temporal IoU. The chain is: frames {ARROW} "
                f"WASB frozen trajectory {ARROW} velocity {ARROW} shuttle-seeded candidate windows {ARROW} "
                "learned scorer weights per-window feature columns {0} graded rallies. The original v0 "
                "segmentation was three hand-tuned knobs (velocity threshold, merge gap, minimum window "
                "duration); the learned scorer replaces that hand-tuning.".format(ARROW)
            )
        )
        story.append(
            P(
                "A key honesty property follows from the data: golden labels are rally <i>times</i> "
                "(when), not (x, y) <i>coordinates</i> (where). They therefore cannot retrain the frozen "
                "WASB detector &mdash; they can only grade and tune the segmentation layer. The discipline "
                "is &ldquo;calibrate on some rallies, measure on held-out ones.&rdquo;"
            )
        )
        story.append(Paragraph("2.2  Signals as feature columns, not branches", H2))
        story.append(
            P(
                "Every producer (detector / analyzer / reconciler) emits a <b>signal = (value, "
                "confidence)</b> persisted to a namespaced column <font face=\"{0}\">"
                "&lt;producer&gt;__&lt;key&gt;</font>. The scoring model consumes those columns and "
                "<b>learns to weight</b> them; there are <b>no hand-coded if/else branches</b> in the "
                "decision path. Producers propose; the scorer arbitrates; the experiment harness and the "
                "golden set decide what is promoted. The scorer itself is deliberately a tiny pure-numpy "
                "L2 logistic regression (ADR-004), <i>not</i> a neural net, so it survives small-N. A cue "
                "that does not earn its keep simply stays at weight ~0 &mdash; it costs nothing because it "
                "is an unused column, not a live branch.".format(FONT_ITALIC)
            )
        )
        story.append(
            P(
                "Fusion has two realizations. The <b>decision-level gate</b> (label-free, ships first) "
```

```
                f"composes Gate A (net-crossings {GE} 2 &mdash; the structural held-shuttle / service-"
                "feed fix) and Gate B (a person cue). The <b>feature-level</b> path (the primary, "
                "learned route) lets cues emit a small set of scale/translation-<i>invariant</i> per-window "
                "features (counts, ratios, zone-membership, two-sidedness &mdash; never raw coordinates, "
                "which would memorize this court and camera) that join the logistic feature vector. The "
                "shuttle stays primary; person is a gate-plus-feature, never the sole trigger, and single-"
                "cue segmenters remain registered fallbacks."
            )
        )

        # ================= 3. Evaluation Methodology (the heart) =================
        story.append(Paragraph("3  Evaluation Methodology", H1))
        story.append(
            P(
                "This section is the heart of the work: the system's credibility rests on <i>how</i> it is "
                "measured, not on any single number."
            )
        )
        story.append(Paragraph("3.1  The owned golden corpus and the oracle contract", H2))
        story.append(
            P(
                "Evaluation uses a small <b>owned golden corpus of n=6 labelled matches</b> "
                "(<font face=\"{0}\">adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, "
                "testlarge_short, mahadevpura_singles</font>; 262 rallies total), grown by the owner, "
                "classified Proprietary, with minors excluded entirely and no biometric/identity labelling. "
                "The labelling vendor, <b>Roora</b>, is treated as an interchangeable <b>oracle label "
                "generator</b>: the system's contract is with the golden <i>labels</i> &mdash; assumed-"
                "trustable ground truth, validated by automated checks plus a ~20% human spot-check at a "
                f"{PM}0.3 s boundary tolerance &mdash; <b>not</b> with their provenance. Evaluation "
                "rigor (interval matching, LOVO, the ship-gate) is therefore <i>decoupled</i> from the "
                "vendor, and the corpus scales by <b>adding or swapping oracles</b>; any oracle's labels "
                "import directly in the same schema. This is the project's moat: a consented dataset plus an "
                "honest eval harness, not the weights.".format(FONT_ITALIC)
            )
        )
        story.append(Paragraph("3.2  Leave-one-video-out, with honest statistics", H2))
        story.append(
            P(
                "Learned variants are scored <b>leave-one-video-out (LOVO)</b>, grouped by match: train on "
                "the <i>other</i> videos' candidates, predict the held-out video, score against its labels. "
                "We never hold out a window from the <i>same</i> video &mdash; windows within a video are "
                "correlated and would flatter the score (the field name is leave-one-
```

```
group/subject-out CV). "
                "Every tunable knob &mdash; the operating-point threshold, the calibrator &mdash; is fit on "
                "the <i>training fold only</i> and never looks at the held-out video. Because a bare LOVO "
                f"mean at n {TIMES if False else GE} 6 is not trustworthy on its own, we attach <b>honest "
                "statistics</b>: a bootstrap 95% confidence interval per metric, a paired "
                f"{DELTA}F1 CI, and an exact paired <b>sign test</b>. The promotion rule is encoded directly: "
                f"a lift is real only when the {DELTA}F1 CI clears 0 <i>and</i> the sign test agrees. This is "
                "necessary because the sign test needs sample size &mdash; at n=6 a ~+0.08 lift cannot reach "
                "p &lt; 0.05; roughly n=10 with 9 wins gives p &asymp; 0.011."
            )
        )
        story.append(Paragraph("3.3  Persist once, re-score forever", H2))
        story.append(
            P(
                "The expensive, non-deterministic per-frame <b>detection</b> (shuttle WASB trajectory, person "
                "tracks, audio hits) runs <b>once</b> per video on the GPU/WSL and is persisted as a small "
                "replayable feature substrate (one JSON per video of resolution-invariant shuttle + person "
                "features per candidate window). The cheap, deterministic <b>decision</b> (windowing, gates, "
                "the logistic scorer) then re-scores any number of variants <b>offline</b>, with no GPU and "
                "zero production risk. A <i>Variant</i> is a declarative re-scoring recipe, not a code branch; "
                "given persisted features it produces the same intervals deterministically. This is what makes "
                "every experiment a zero-risk offline re-score &mdash; the reproducibility thesis of the whole "
                "harness."
            )
        )

        story.append(PageBreak())

        # ================= 4. Results =================
        story.append(Paragraph("4  Results", H1))
        story.append(Paragraph("4.1  Baselines that drive strategy", H2))
        story.append(
            P(
                "Table 1 anchors strategy. Two reference footages are reported: <b>Mahadevpura</b> "
                "(clean-ish, single court) and <b>Kushagra</b> (multi-court, the hard case, contrast "
                f"1.4{TIMES}). The reading is blunt: clean footage is commoditized (a Gemini serve reaches "
                "0.93), but multi-court drops the cloud wrapper to <b>0.66</b> &mdash; and that 0.66 is the "
                "owned model's ship target. The owned Gen-0 head already reaches 0.561 multi-court at n=6."
            )
        )
        tbl1 = [
            ["Method", "Mahadevpura (clean)", "Kushagra (multi-court)"],
            ["Heuristic (velocity windowing)", "0.657", "0.157"],
            ["Gen-0 owned head (LOVO, n=6)", "0.744", "0.561"],
            ["Two-stage WASB + Gemini refine", "0.83", "&mdash; (not measured)"],
            ["Gemini wrapper (gemini-flash-latest)", "0.93", "0.66"],
```

```
        ]
        story.append(make_table(tbl1, [7.2 * cm, 4.65 * cm, 4.65 * cm], align_centers=[1, 2]))
        story.append(
            Paragraph(
                "Table 1. Rally-detection F1 at IoU 0.5 vs human golden labels. "
                "Clean footage is commoditized; multi-court 0.66 is the owned-model ship target.",
                CAPTION,
            )
        )
        story.append(
            P(
                "Per-video detail on the cloud reference (gemini-flash-latest): on Mahadevpura, F1 0.93, "
                "precision 1.00, recall 0.87, mean IoU 0.80, boundary error 0.7 s (n_gt 31, n_pred 27; "
                "3 of 4 false negatives were detected by Gemini but rejected by a 3.0 s minimum-segment "
                "filter on 2.5 s rallies). On the harder Kushagra, F1 0.66, precision 0.69, recall 0.62, "
                "mean IoU 0.73, boundary error 1.2 s (n_gt 32, n_pred 29)."
            )
        )

        story.append(Paragraph("4.2  Gate-A: the first honestly-promoted cue", H2))
        story.append(
            P(
                f"Gate A (net-crossings {GE} 2) is the first cue promoted through the harness, because it "
                "is data-backed: real rallies on multi-court matches cross the net 2 or more times <b>97% of "
                "the time</b> (median 5), whereas a single service feed crosses about once. The measured "
                "result on the recall-oriented proposal candidate set is unambiguous (Table 2)."
            )
        )
        tbl2 = [
            ["Metric", "Result"],
            [f"{DELTA}F1 (Gate-A vs control)", "<b>+0.074</b>"],
            ["Golden matches won", "6 / 6"],
            ["Paired sign-test p", "<b>0.031</b>"],
            [f"95% CI on {DELTA}F1", "[+0.042, +0.104]  (clears 0)"],
            ["Over-segmentation (segment-count ratio)", f"1.68 {ARROW} 1.01"],
            ["Recommended config", "<font face=\"%s\">indexing.fusion.min_crossings = 2</font>" % FONT_ITALIC],
        ]
        story.append(make_table(tbl2, [8.3 * cm, 8.2 * cm], align_centers=[1]))
        story.append(
            Paragraph(
                "Table 2. Gate-A promotion statistics (2026-06-14). The CI clears 0 and the sign test agrees "
                "? the only fusion cue to date that does.",
                CAPTION,
            )
        )
        story.append(
            P(
                "Real-footage corroboration (no golden labels, so this is a divergence check, not an F1): an "
                "offline re-score of GX010125_1080p's cached trajectory took 45 candidates from CONTROL's 45 "
                "rallies down to Gate-A's 27, dropping 18 low-crossing windows
```

```
(~1.7 min) of likely "
                "held-shuttle / feed false positives. Caveat: this offline win is on the recall-oriented "
                "proposal set; flipping the live served default remains owner-gated, since the production "
                "fusion segmenter is still mock-tested rather than verified on real video."
            )
        )

        story.append(Paragraph("4.3  The learned-variant fix (PR #130)", H2))
        story.append(
            P(
                "An early end-to-end A/B of a window-level 5-feature logistic variant <i>lost</i> to control "
                f"({DELTA}F1 = &minus;0.081, 2W/4L) and zeroed the testlarge video &mdash; but this was a "
                "fixed-0.5-cutoff artifact, not a real defect of the stronger per-frame prototype. PR #130 "
                "added an additive, default-OFF threshold-tuning step (operating point chosen on the train "
                "fold) plus probability calibration (Platt / isotonic). Validated on the n=6 golden set, mean "
                "F1 rose <b>0.307 &rarr; 0.423 (+0.116)</b>, testlarge recovered from 0.000 to 0.300, and the "
                "learned variant now beats CONTROL (0.388). The lesson &mdash; calibrate and tune the "
                "operating point <i>inside</i> the LOVO loop &mdash; is now baked into the harness."
            )
        )

        story.append(Paragraph("4.4  Fusion ablation: person (P3) and audio (P4)", H2))
        story.append(
            P(
                "Two further cues were measured incrementally on the same n=6 corpus (honest LOVO, IoU 0.5). "
                "Neither clears the promotion bar &mdash; both CIs span 0 and both sign tests are non-"
                "significant &mdash; so both stay <b>default-OFF</b>. We report them in full because negative "
                "and uncertain results are the honest ablation backbone (Table 3, Figure 1). Values "
                "below use the full-precision leaderboard floats; the doc rounds them."
            )
        )
        tbl3 = [
            ["Step", "F1", f"{DELTA}F1 vs prior", "95% CI", "Sign test", "Clears 0?"],
            ["shuttle-only", "0.3069", "&mdash;", "&mdash;", "&mdash;", "&mdash;"],
            ["+person (P3)", "0.2863", "&minus;0.0205 (&minus;6.7%)", "[&minus;0.165, +0.162]", "2W/4L, p=0.6875", "No"],
            ["+person+audio (P4)", "0.3656", "+0.0792 (+27.7%)", "[&minus;0.054, +0.203]", "4W/2L, p=0.6875", "No"],
        ]
        story.append(
            make_table(
                tbl3,
                [3.0 * cm, 1.7 * cm, 3.2 * cm, 3.0 * cm, 2.7 * cm, 1.9 * cm],
                align_centers=[1, 2, 3, 4, 5],
                font_size=7.6,
            )
        )
        story.append(
            Paragraph(
                "Table 3. Fusion ablation (n=6, honest LOVO, IoU 0.5; full-precision floats
```

```
from "
                "experiment_leaderboard.json). Both records: IoU 0.5, label_set_hash 50d98ffbc370, n=6.",
                CAPTION,
            )
        )
        story.append(
            P(
                "Person adds noise: without court polygons, the players-in-court count includes spectators, "
                "so the cue is a wash (per-video: testlarge +0.40, gbaaddy &minus;0.22). Audio is the more "
                "promising lead &mdash; it wins 4/6 (largetest +0.27, mahadevpura +0.27) and lifts F1 to "
                "0.366, above the 0.307 shuttle-only baseline &mdash; but n=6 simply cannot confirm it. The "
                "discipline holds: promote only on a lift whose CI clears 0."
            )
        )
        # Figure 1
        if os.path.exists(CHART_PATH):
            img = Image(CHART_PATH, width=15.2 * cm, height=8.06 * cm)
            story.append(Spacer(1, 0.1 * cm))
            story.append(img)
            story.append(
                Paragraph(
                    "Figure 1. Incremental fusion ablation on the n=6 golden corpus. Neither cue's "
                    "?F1 confidence interval clears zero; both remain default-OFF. Dashed line: the "
                    "heuristic LOVO floor (0.480) the learned scorer must beat.",
                    CAPTION,
                )
            )

        # ================= 5. Related Work =================
        story.append(Paragraph("5  Related Work (cite, don't claim)", H1))
        story.append(
            P(
                "A literature review (multi-agent web search, every citation adversarially verified &mdash; "
                "zero fabricated papers across ~40 checks) settles what is standard versus novel. The "
                "architecture's <b>skeleton is standard and must be cited, not claimed</b>. "
                "Candidate-windows-then-learned-scorer is <i>temporal action localization</i> (TAL, "
                "proposal-then-score). The reconciler's over-segmentation fixes are <i>temporal action "
                "segmentation</i> (TAS) refinement / post-processing. &ldquo;Signal not verdict / a learned "
                "arbiter over cue columns&rdquo; is late / score-level fusion realized as stacked "
                "generalization &mdash; roughly 20 years old and already present in racket sports (CVIU 2009; "
                "Ghosh et al. 2018 SVM-over-features). In-domain prior art (See et al. 2024/25 rally start/end; "
                "MonoTrack; Hsu et al. 2024; Goh et al. 2023 service fault; ShuttleSet) hard-codes the same "
                "cues as if/else verdicts."
            )
        )
        story.append(
            P(
                "Against that backdrop, <b>what is genuinely novel</b> is narrow and we park
```

```
our claims there: "
                 "(1) the <b>report-first reconciler-as-linter</b> bundle &mdash; contradiction-detection over a "
                 "persisted temporal decision, with auditable structured interval corrections, per-rule A/B "
                 "lift-gating <i>before</i> auto-apply, idempotency and fixed precedence, human-in-the-loop, and "
                 "feeding a (value, confidence) scorer &mdash; for which no surveyed paper bundles all parts; and "
                 "(2) this <b>disciplined small-N promotion gate</b> on a deliberately tiny golden set. "
                 "Everything else (the fusion mechanism, the persist-once substrate, cross-check-and-correct, "
                 "the no-regression harness) is textbook and we claim only the specific discipline, not the idea."
        )
    )

    # ================= 6. Limitations =================
    story.append(Paragraph("6  Limitations", H1))
    lims = [
        ("Sample size n=6.", "Enough for direction-checks and LOVO sign-tests, not enough to claim a "
         "general model. At n=6 a ~+0.08 lift cannot reach p &lt; 0.05; corpus growth is the highest-ROI "
         "unblock."),
        ("CIs span 0 for person and audio.", "Both fusion cues (P3, P4) have ?F1 confidence "
         "intervals straddling zero and non-significant sign tests, so neither is promotable yet despite "
         "audio's encouraging +27.7% point estimate."),
        ("Court polygons absent.", "Without per-video court masks the person cue counts spectators, "
         "which is the mechanical reason +person is a wash. Activating court polygons is the obvious next "
         "fix."),
        ("Served path is mock-tested.", "The production fusion segmenter is verified offline on cached "
         "trajectories, not yet on live real video; flipping the served default stays owner-gated."),
        ("Detector domain gap unmeasured.", "WASB/TrackNet were trained on other rigs; detector accuracy "
         "on our footage is not yet quantified, and every downstream cue inherits that gap."),
    ]
    for head, txt in lims:
        story.append(P(f"<b>{head}</b> {txt}", BODY_TIGHT))

    # ================= 7. Planned Work =================
    story.append(Paragraph("7  Planned Work", H1))
    plans = [
        ("Grow the corpus via the oracle.", "The #1 lever. Priority: multi-court badminton (the target "
         "distribution and where the ship target lives), then &ge;3 matches per new sport (table tennis "
         "first &mdash; WASB transfers at F1 0.909). At n=10 with 9 wins, p &asymp; 0.011 &mdash; "
         "significance is reachable within a month, adults only."),
        ("Activate court polygons.", "The intake manifest already collects court_corners, net_line, and "
         "singles/doubles per video, so the detector can ignore people outside the court and require &ge;2 "
         "players on court &mdash; the direct fix for the person-cue noise above."),
        ("Measure audio on shuttle-only directly.", "Audio is parked (E1 probe found ~2.2× "
```

```
             "discrimination, AUC ~0.6), not in reserve; the planned Fusion-P4 path lands it flag-gated and "
             "default-OFF, promoted only on a measured LOVO lift."),
            ("Shuttle-dropped cue + wire Gate-A to production.", "The cheapest current win is wiring the "
             "already-built net-crossing Gate-A into the live segmenter (it is presently eval-only). A faithful "
             "serve-contact boundary definition structurally fixes the held-shuttle false positive; a future "
             "&ldquo;shuttle-on-floor&rdquo; cue extends the same substrate reasoning."),
            ("Owned model: Qwen3-VL (Apache-2.0).", "The roadmap starts with the TAL head, not the VLM, "
             "because boundary accuracy is the product and fine-tuned VLMs miss strict boundaries. Base model "
             "Qwen3-VL (Apache-2.0), InternVL3 (MIT) fallback; commercial-clean licensing for code, data, and "
             "weights; never train on paid-LLM outputs."),
            ("Execution-policy resilience.", "Motivated by a real Gemini hang with no client-side timeout, a "
             "per-job declarative ExecutionPolicy (op_timeout_sec, max_wait_sec, self-scheduling via the DB as "
             "coordinator) makes every long operation declare how it waits, aborts, and resumes."),
        ]
        for head, txt in plans:
            story.append(P(f"<b>{head}</b> {txt}", BODY_TIGHT))

        # ================= 8. Engineering & Reproducibility =================
        story.append(Paragraph("8  Engineering &amp; Reproducibility", H1))
        story.append(
            P(
                "<b>Telemetry black box.</b> A rotating ~10 KB JSONL runlog (atomic tmp + os.replace on "
                "every write, so a hard kill never leaves it half-written) records execution velocity plus "
                "machine health, so a post-mortem on an OS/OOM/thermal kill with no Python traceback is a "
                "bullseye instead of a guess. Each sample captures GPU temp/util/power/clocks/VRAM (with a "
                "derived throttle flag at SM clock &lt; 0.8&times; max or temp &ge; 84 C) and CPU/RAM. "
                "It is the observability companion to the ExecutionPolicy control layer."
            )
        )
        story.append(
            P(
                "<b>Experiment leaderboard.</b> Each compare() appends one compact reproducible row to "
                "<font face=\"%s\">eval_baselines/experiment_leaderboard.json</font> &mdash; a tracked location "
                "that survives a clean checkout. Every row is pinned to (exact recipe via a 12-char "
                "spec_hash, exact label set via a label_set_hash, exact IoU), so any regression is traceable "
                "to a recipe change rather than drift." % FONT_ITALIC
            )
        )
        story.append(
            P(
                "<b>CI gates.</b> Four gates run on every PR (Python 3.13): Ruff lint over the whole repo; a "
                "lenient-but-green mypy baseline with an explicit ratchet-down quarantine; a syntax + import "
                "smoke with no GPU stack (guarding the class of breakage where a module-level import error "
```

```
                "ships while the suite stays green); and the full offline mocked suite with a coverage floor "
                "(&minus;&minus;cov-fail-under=70, calibrated against the measured 72%). Combined with default-"
                "OFF cues and a pinned CONTROL variant, this makes rollback a config flip, never a git revert."
            )
        )

        # ================= Conclusion =================
        story.append(Paragraph("9  Conclusion", H1))
        story.append(
            P(
                "The contribution of this work is less any single F1 number and more a <b>method for being "
                "honest at small scale</b>. On a 6-match owned golden corpus, a shuttle-trajectory substrate "
                "feeding a tiny logistic scorer, with every cue entering as a default-OFF feature column and "
                "promoted only on a held-out lift whose confidence interval clears zero, produced exactly one "
                f"honestly-promoted cue ({DELTA}F1 +0.074, p=0.031) and two cues that &mdash; reported in full "
                "&mdash; do not yet clear the bar. By treating the labelling vendor as a swappable oracle, the "
                "evaluation rigor is decoupled from data provenance, so the cheapest path to a publishable "
                "result is simply more labelled matches: at n=10 with 9 wins, significance is one month away. "
                "The architecture is deliberately standard where the literature is mature and claims novelty "
                "only on the report-first reconciler and the small-N promotion discipline &mdash; the honest "
                "ablation backbone this report is built to be."
            )
        )

        return story


    def main():
        build_chart()

        doc = BaseDocTemplate(
            PDF_PATH,
            pagesize=A4,
            leftMargin=2 * cm,
            rightMargin=2 * cm,
            topMargin=1.8 * cm,
            bottomMargin=1.8 * cm,
            title="Local, Cheap, AI Rally Detection for Racket Sports - Quality Report",
            author="badminton-highlight-indexer",
        )
        frame = Frame(
            doc.leftMargin,
            doc.bottomMargin,
            doc.width,
            doc.height,
            id="body",
            leftPadding=0,
            rightPadding=0,
            topPadding=0,
            bottomPadding=0,
        )
        first = PageTemplate(id="First", frames=[frame], onPage=on_first_page)
```

```
    later = PageTemplate(id="Later", frames=[frame], onPage=on_page)
    doc.addPageTemplates([first, later])

    story = build_story()
    # Page 1 uses the banner template (First); every page after switches to the
    # running-header template (Later). NextPageTemplate takes effect at the next
    # page break, so it must be emitted on page 1.
    story.insert(0, NextPageTemplate("Later"))
    # Banner occupies the top 4.4 cm; top margin is 1.8 cm, so push the title
    # block down ~3.0 cm to clear the navy banner on page 1.
    story.insert(1, Spacer(1, 3.0 * cm))

    doc.build(story)

    size = os.path.getsize(PDF_PATH)
    print(f"Wrote {PDF_PATH} ({size} bytes)")
    # cleanup the temp chart? keep it for inspection
    return size


if __name__ == "__main__":
    main()

```

4. assistant (2026-06-14T07:01:50.376Z)

I'll make surgical edits to restyle this report for an academic-journal look. Let me start with the font registration and color palette.

5. user (2026-06-14T07:02:09.470Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py has been updated successfully. (file state is current in your context ? no need to Read it back)

6. assistant (2026-06-14T07:02:12.430Z)

Now the color palette. Drop crimson, keep a single navy accent, and remove the green/red number-coloring colors.

7. user (2026-06-14T07:02:17.944Z)

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8. assistant (2026-06-14T07:02:21.080Z)

Now update the styles. Body to 10pt/13.5 leading, headings navy, subtitle to black italic, table header to bold black, captions serif.

9. user (2026-06-14T07:02:37.640Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py has been updated successfully. (file state is current in your context ? no need to Read it back)

10. assistant (2026-06-14T07:02:41.117Z)

Now the matplotlib chart ? muted grey/navy bars, black labels, dashed grey floor line, no bright colors, >=150 dpi.

11. user (2026-06-14T07:02:59.254Z)

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12. assistant (2026-06-14T07:03:04.578Z)

Now the table builder ? booktabs style: thin top rule, bold black header on light grey, thin rule under header, horizontal hairlines only, thin bottom rule, very faint alternating shading.

13. user (2026-06-14T07:03:16.102Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py has been updated successfully. (file state is current in your context ? no need to Read it back)

14. assistant (2026-06-14T07:03:20.169Z)

Now the key-result strip table (uses STEEL/LIGHT and an INNERGRID). Let me restrain it to match the booktabs aesthetic.

15. user (2026-06-14T07:03:26.857Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py has been updated successfully. (file state is current in your context ? no need to Read it back)

16. assistant (2026-06-14T07:03:30.012Z)

Now the title page and running header functions. Remove the navy banner + red strip; clean academic first page. Let me update the footer, the running header, and the first-page function.

17. user (2026-06-14T07:03:44.692Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py has been updated successfully. (file state is current in your context ? no need to Read it back)

18. assistant (2026-06-14T07:03:48.815Z)

Now I need to fix the title-block spacer in `main()` ? it was pushing the title down 3.0cm to clear the (now-removed) banner. Let me reduce it to a small top margin.

19. user (2026-06-14T07:03:51.722Z)

The file C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\quality_report.py has been updated successfully. (file state is current in your context ? no need to Read it back)

20. assistant (2026-06-14T07:03:54.942Z)

Now let me check for any remaining references to deleted constants (ACCENT, HEADER_BG, GREEN, STEEL) that would break the script.

21. user (2026-06-14T07:03:55.854Z)

```
150:STEEL = NAVY  # subheadings reuse the same navy (no second accent)
386:        ("ROWBACKGROUNDS", (0, header_rows), (-1, -1), [colors.white, ROW_ALT]),
```

22. assistant (2026-06-14T07:03:59.058Z)

No dangling references to deleted `ACCENT`/`HEADER_BG`/`GREEN`. STEEL is aliased to NAVY (harmless) and `colors.white` is a valid reportlab color used as a row background. Let me also confirm there's no conditional green/red number coloring in the table cells (e.g. via inline `<font color>` tags) that I missed.

23. user (2026-06-14T07:04:00.552Z)

No matches found

24. assistant (2026-06-14T07:04:03.363Z)

All the old hex colors and conditional coloring are gone. Now let me run the report generator.

25. user (2026-06-14T07:04:05.457Z)

Wrote C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\Rally_Detection_Quality_Report.pdf (256383 bytes)

26. assistant (2026-06-14T07:04:11.378Z)

It ran. Let me confirm the regenerated PDF with `ls -la` and render all pages to PNG.

27. user (2026-06-14T07:04:12.939Z)

```
-rw-r--r-- 1 avidu 197609 256383 Jun 14 12:34 scratch/Rally_Detection_Quality_Report.pdf
-rw-r--r-- 1 avidu 197609  57450 Jun 14 12:34 scratch/_fusion_ablation_chart.png
```

28. user (2026-06-14T07:04:14.647Z)

pages: 7

29. assistant (2026-06-14T07:04:17.681Z)

7 pages rendered. Let me visually review the key pages ? the title page, a table page, and the figure page ? to confirm the conservative academic look.

30. assistant (2026-06-14T07:04:21.856Z)

Title page looks clean and academic: serif Times body, centered navy bold title, black italic subtitle, no banner, booktabs key-result strip. Let me check the figure page and a results table page.

31. assistant (2026-06-14T07:04:25.935Z)

The figure page looks good: muted navy/grey bars, black labels, dashed grey floor line, no saturated colors; the booktabs table (Table 3) has a thin top rule, bold-black header on light grey, hairline body rules, thin bottom rule, and all numbers plain black (no green/red). Let me verify one more table page and the running header on page 2.

32. assistant (2026-06-14T07:04:29.605Z)

Both Table 1 and Table 2 render in the restrained booktabs style with the grey running header at top, plain-black numbers, and thin navy/black rules. The restyle is complete and verified across pages.

Let me clean up the temporary review PNGs.

33. user (2026-06-14T07:04:31.090Z)

cleaned review PNGs

34. assistant (2026-06-14T07:04:38.864Z)

Done. Confirmation: `scratch/quality_report.py` ran cleanly and regenerated the PDF.

PDF: `C:\Users\avidu\Projects\badminton-highlight-indexer\scratch\Rally_Detection_Quality_Report.pdf` ? 256,383 bytes (~250 KB), 7 pages. Verified visually across the title page, both table pages, and the figure page (rendered to PNG at 120 dpi, then removed).

What changed (style only ? no text/numbers touched):
1. Switched body/abstract/captions/tables to serif Times New Roman (registered all 4 variants via TTFont + registerFontFamily, Arial?Helvetica fallback chain), body now 10pt/13.5 leading justified; collapsed the palette to one navy accent (#1F3A5F) with everything else plain black,

deleting the crimson accent and all green/red conditional number coloring.
2. Rebuilt tables in booktabs style (thin top/header/bottom rules, bold-black header on faint #F0F0F0 fill, hairline body rows, no vertical grid) and replaced the title page's navy banner + red strip with a clean centered serif title + black italic subtitle; the matplotlib figure now uses muted navy/grey bars, black labels, a dashed grey floor line, saved at 200 dpi.